

Zhihao Chen

aa2212448547@gmail.com | github.com/zhihaochen67 | zhihaochen67.github.io | Nanjing, China

EDUCATION

Nanjing University of Posts and Telecommunications (NJUPT)

Sep 2024 - Jun 2028

B.S. in Computer Science | Expected graduation Jun 2028 | CET-6

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks

EXPERIENCE

Institute of Industrial Artificial Intelligence, CAS (IAII) - Research Intern

2026 - Present

- Work on embodied AI, social navigation, and robotic manipulation, building reproducible simulation and evaluation pipelines with an emphasis on safety and closed-loop verification.
- Built benchmarked PyBullet systems: predictive navigation shielding reduced collision rate from 18% to 0% (N=3), 23% to 1% (N=5), and 43% to 7% (N=10); bounded manipulation recovery reached 96/100 grasp and 100/100 placement success.

SELECTED PROJECTS

AgentLab - LLM Agent Evaluation & Observability

- Built reproducible LLM-agent evaluation infrastructure with isolated workspaces, deterministic pytest/file-change gates, structured traces, SQLite persistence, repeated trials, experiment comparison, and optional LLM-as-a-Judge.
- Added resumable evaluation for approval-gated agents while keeping deterministic verification authoritative; persisted evaluator outcomes, latency, failure taxonomy, and replayable traces.

MCP ToolHub - Secure MCP Execution Gateway

- Built a local stdio-only MCP gateway exposing exactly 14 production tools across filesystem, shell, Git, audit, and control surfaces.
- Implemented out-of-band human approval, immutable expiring single-use requests, replay-resistant execution, workspace confinement, Contract V1 structured outcomes, and trace-linked auditability.

Repo Doctor - AI Repository Diagnosis & Repair

- Built a local-first CLI combining deterministic repository diagnostics with opt-in LLM analysis, constrained patch generation, verification, exact-byte rollback, and optional MCP execution.
- Validated v0.3.0 with 227 tests passed, 5 skipped and a 100/100 self-scan; added approval-aware MCP suspend/resume for protected repair operations.

HybridSearch-Bench - Retrieval & Reranking Benchmark

- Implemented BM25 from scratch, dense retrieval with Sentence Transformers + FAISS, weighted/RRF fusion, cross-encoder reranking, and reproducible evaluation across SciFact and FiQA.
- Reached 0.6998 nDCG@10 on SciFact with hybrid weighted retrieval + rerank@20 (BM25 0.5694; dense 0.6451; hybrid 0.6771) while measuring quality-latency trade-offs.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript

AI / Agents: PyTorch, Transformers, LLM Agents, MCP, Agent Evaluation, LLM-as-a-Judge

Search / Retrieval: BM25, Dense Retrieval, Hybrid Search, Reranking, FAISS

Engineering: Linux, Git, SQLite, pytest, Docker

Robotics: PyBullet, RGB-D, Social Navigation, Robotic Manipulation